Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

Illustrate the Lloyd-Max algorithm for scalar quantization, assuming full knowledge of the source pdf. How does the rate-distortion performance of Lloyd-Max compare with other conventional quantizers (uniform, ECSQ, VQ)?
Explain how the Lloyd-Max quantizer can be designed when the pdf of the source is unknown but a training set of N samples is given.

- Write the pseudocode of the algorithm in this case
- Simulate the execution of the algorithm generating $N = 4$ samples at random in the interval $[0, 1]$.

2 Question - 5 pts

Illustrate the basic principles behind the MDCT. Illustrate how the MDCT is able to accommodate signals with non-stationary time-frequency characteristics.

3 Question - 12 pts [Matlab]

Consider a memoryless source, uniformly distributed in the interval $[-1, 1]$.

- Design a suitable quantizer for the source at hand.
- For each target bitrate R [bps], *empirically* measure the output distortion of the uniform scalar quantizer.
- Plot the rate-distortion functions of the quantizer.
- Compare the plots with the theoretical Shannon's bound.

4 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a discrete memoryless source with finite alphabet, and $D(R)$ the corresponding rate-distortion curve
   - [T] [F] $D(0) = H(X)$
   - [T] [F] If $\langle D, R\rangle$ is achievable, than $\langle D, R + \Delta R\rangle$, $\Delta R > 0$ is achievable
   - [T] [F] $R(0) = \infty$
   - [T] [F] $H(X) \leq \log_2(M)$, where M is the alphabet size
2. Consider a speech coding system
   - [T] [F] The long term predictor is useful when coding unvoiced speech
   - [T] [F] Direct quantization of prediction coefficients might lead to instability at the decoder
   - [T] [F] The number of poles of the whitening filter $A(z)$ depends on the number of formants below 4 kHz
3. Consider a video coding system. Given a video sequence of fixed spatio-temporal resolution:
   - [T] [F] The computational complexity of motion estimation does not depend on the block size.
   - [T] [F] The coding efficiency increases when increasing the GOP size (e.g. the time interval between I-frames).
   - [T] [F] Entropy coding is the most computationally intensive operation.